

2020 Test 1 solution

Nanyang Technological University
SPMS/Division of Mathematical Sciences

AY20/21 MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS Test 1

Name &
Matriculation Number

Suggested Solutions

Score:

Venue:

TIME: 9:35 -10:15, 15 September, 2020

1. (40 Points) A factory finds that, on average, 20% of the bolts produced by a given machine will be defective for certain specified requirements. If 10 bolts are selected at random from the day's production of this machine, find the probability
- (a) that exactly 2 will be defective;
 - (b) that 2 or more will be defective.

(a) let X be the r.v. denoting the number of bolts selected that are defective.

$$X \sim \text{Bin}(10, 0.2)$$

$$\begin{aligned} P(X=2) &= \binom{10}{2} (0.2)^2 (1-0.2)^8 \\ &\approx 0.302, \end{aligned}$$

$$\begin{aligned} (b) \quad P(X \geq 2) &= 1 - P(X \leq 1) \\ &= 1 - [P(X=0) + P(X=1)] \\ &= 1 - \left[\binom{10}{0} (0.2)^0 (1-0.2)^{10} + \binom{10}{1} (0.2)^1 (1-0.2)^9 \right] \\ &= 1 - 0.375896 \\ &\approx 0.624, \end{aligned}$$

2. (30 Points) A laboratory blood test is 95% effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1% of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5% of the population actually has the disease, what is the probability that a person has the disease, that the test result is positive?

given

Let D be the event that a person has a certain disease.

Let T be the event that the test shows a positive result.

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D')P(D')} \quad (\text{Bayes' Rule})$$

$$P(T|D) = 0.95, \quad P(T|D') = 0.01, \quad P(D) = 0.005$$

$$\begin{aligned} \therefore P(D|T) &= \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(1-0.005)} \\ &\approx 0.323, \end{aligned}$$

3. (30 Points) The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ . However, such a random variable can be observed only if it is positive, since if it is 0, then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

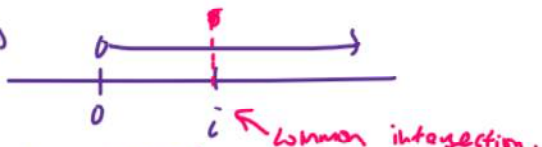
$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ . Find $E(Y)$.

$X \sim \text{Poisson}(\lambda)$

$$P\{Y=i\} = P\{X=i | X>0\}$$

$$= \frac{P(\{X=i\} \cap \{X>0\})}{P(X>0)} \quad (\text{conditional probability})$$

$$= \frac{P\{X=i\}}{1 - P(X=0)} \rightarrow \text{complement}$$


$$= \frac{e^{-\lambda} \cdot \frac{\lambda^i}{i!}}{1 - e^{-\lambda}}$$

$$E(Y) = \sum_{i=1}^{\infty} P\{Y=i\} \quad (\text{As } Y>0)$$

$$= \sum_{i=1}^{\infty} i \left[\frac{e^{-\lambda} \cdot \frac{\lambda^i}{i!}}{1 - e^{-\lambda}} \right]$$

$$= \frac{1}{1 - e^{-\lambda}} \sum_{i=1}^{\infty} i \cdot \frac{\lambda^i}{i!} \quad Y>0$$

$$= \frac{1}{1 - e^{-\lambda}} \sum_{i=1}^{\infty} \frac{\lambda^i e^{-\lambda}}{(i-1)!}$$

$$\begin{aligned} & \text{let } j = i-1 \\ & \text{then } i=1, j=0 \\ & = \frac{1}{1 - e^{-\lambda}} \sum_{j=0}^{\infty} \frac{\lambda^{j+1} e^{-\lambda}}{j!} \\ & = \frac{\lambda}{1 - e^{-\lambda}} \left[\sum_{j=0}^{\infty} \frac{\lambda^j}{j!} e^{-\lambda} \right] \\ & = \frac{\lambda}{1 - e^{-\lambda}} \cdot 1 \quad (\text{sum of Poisson prob}) \end{aligned}$$

0-truncated Poisson variable.